

Low-Cost Modular Planar Near-Field Scanner

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Abstract—Planar near-field scanners provide a simple method of characterizing radiation. They have a long history for determining both near- and far-field radiation patterns. Additionally, they have a form factor which is flexible and compatible with a laboratory setting. However, the cost of planar near-field scanners can be prohibitive for small labs and hobbyists. Here we present a low-cost open-source planar near-field scanner that can be built from readily available components. The design is modular, allowing for radiation measurements tailored to a broad range of form-factors and application spaces. The accuracy of the scanner is verified by comparing measured data of a standard gain horn to full-wave simulations. The low cost of the scanner and public access to the design reduces the barrier to entry for radiation measurements, expanding this capability to small research groups, small businesses, and hobbyists.

Index Terms—Affordable, antenna measurement, low-cost, microwave measurement, near-field scanner, open-ended waveguide probe, planar near-field measurement.

I. INTRODUCTION

Planar near-field scanners provide a simple and reliable tool to conveniently determine the radiation characteristics of a device. Planar scanners can measure both near- and far-field radiation characteristics in an indoor, environmentally controlled setup. With respect to far-field chambers, near-field scanners are compact and thus compatible with a small laboratory space. Tabletop near-field scanners enable in-house radiation measurements in almost any workspace, opening up design and diagnostic workflows by providing immediate radiation feedback. The simplicity of planar scanners extends to post processing, where far-field transforms can be calculated using the two dimensional fast Fourier transform. A planar near-field scanner is convenient to install because of its size, to operate because of its simplicity, and to use because of the straightforward post processing. The planar near-field scanner presented in this paper makes these benefits accessible by reducing the cost and providing a design reference for the reader to construct their own near-field scanner.

The planar near-field scanner presented in this paper (shown in Fig. 1) supports modest far-field accuracy for high directivity radiators at a price point of approximately \$2,000. The design of the scanner draws upon other reported implementations reported within the literature [1]–[4] as well as classical antenna theory [5]. For the mechanical motion, the scanner design leverages advances in the open-source computer numerical control (CNC) hobbyist community for a mature and robust foundation for the software and hardware of the design. Because the design of the scanner has been open sourced [6], the reader is free to improve and tailor the design for their own application space. In addition, the core

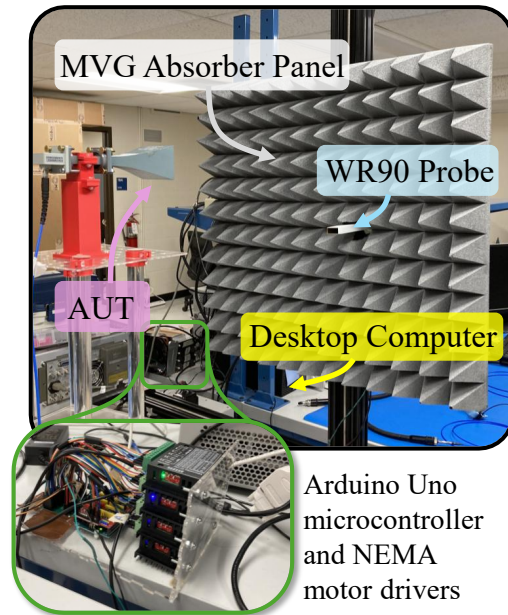


Figure 1: Photograph of the developed planar near-field scanner configured to measure the Narda 640 standard gain horn antenna. The developed near-field scanner is an ideal form-factor for a small laboratory setting.

functionality is modular, making it quick to switch between measurement configurations and straightforward to repair and upgrade core components of the scanner. The design is also relatively simple, making it easier to assemble and understand.

II. SYSTEM WALKTHROUGH

The scanner is made up of four core components: the computer, which provides centralized control for the system; the vector network analyzer (VNA), which measures the transmission amplitude and phase from the antenna to the probe; a microcontroller (an Arduino Uno in the realized system), which provides motor control; and the mechanical assembly, which comprises all the mechanical parts of the scanner, and moves the probe relative to the antenna under test (AUT) in a repeatable and reliable manner.

The computer is the main interface for the user. Using a MATLAB graphical user interface (GUI), the user can both manually control the near-field scanner as well as run parameterized planar near-field measurements. The GUI includes functions that guide the user to calibrate the machine for a given measurement setup. Finally, the GUI includes a function

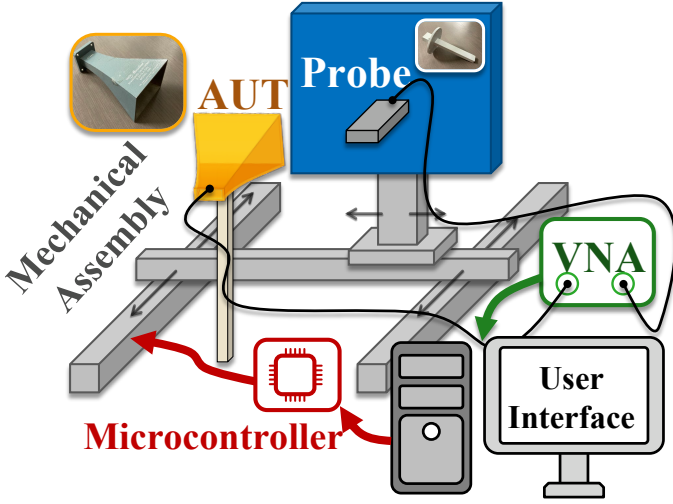


Figure 2: A visual overview of the planar near-field scanner. The primary components are the computer running MATLAB, Arduino microcontroller running grbl, the VNA taking measurements, and the mechanical assembly which positions the probe accurately relative to the AUT.

to plot near-field results and a time gating GUI to visually time gate results.

The computer interfaces with an Arduino over a serial connection to control the mechanical assembly. To translate computer commands to motor driver signals, the Arduino has been flashed with the open-source motor control software grbl [7] and upgraded with a CNC hat. The two main functions carried out by grbl are the homing and the motor movement. Through limit switches on each primary axis, the scanner homes before every measurement. Homing provides absolute positioning to the scanner, allowing a measurement setup to be reused between reboots and for software safety features that prevent the user from pushing the scanner out of bounds. The grbl software provides ramp up and down speed control to the stepper motors to increase repeatability and reduce strain on the motors and fasteners of the scanner. The theoretical resolution of the linear movement is 10 ± 1 microns. However, the practical resolution is closer to 1 ± 0.1 millimeters. Because the design uses stepper motors, the positional accuracy extends through the entire range of motion.

The computer communicates with a VNA (in our case the PNA N5224A) to send commands to take an S parameter measurement and to receive the S parameter data. The VNA is connected to the AUT, which is stationary in the near-field scanner, and the probe, which moves in a raster pattern. Measurements are taken after the probe has been repositioned by the mechanical assembly. Though the full S2P file is saved for each position, only S_{21} is used in post processing.

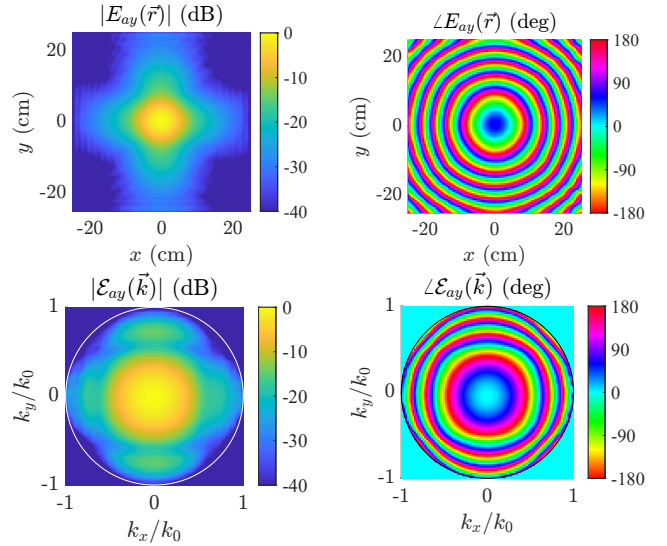


Figure 3: Probe-compensated spatial-domain (top row) and spectral-domain (bottom row) amplitude and phase computed using the setup in Fig. 1. The data has been probe-compensated using (3) and time-gated to mitigate the impact of multiple reflections. Note that only the $\hat{y}$ polarization is measured.

III. PROCESSING AND VALIDATION

A. Measurement Setup

To verify the scanner measurement results align with simulation, a Narda 640 standard gain horn was 3D modeled, simulated, and measured. To 3D model the Narda horn, a pair of calipers and basic manual photogrammetry were used, with particular focus on the dimensional accuracy of the aperture and WR90 input port. For the simulation, the 3D model was simulated in Ansys High Frequency Structure Simulator (HFSS). For the measurement, the scan parameters were a 500 mm scan width, a 120 mm distance between the probe and antenna, and a 14.7 mm sampling distance between points. The VNA was set to 15 dBm output power for maximum signal to noise ratio (SNR) and connected to the Narda horn and WR90 open-ended waveguide probe.

B. Theory and Post Processing

The procedure used for performing the probe compensation as well as computing the far-field from the aperture field distribution is adapted from [5], [8], [9]. Define the spatial spectrum of the measured complex transmitted signal, $S(x, y)$, from the AUT to the probe as

$$\tilde{S}(\nu_x, \nu_y) = \iint_{\text{Scan Area}} S(x, y) e^{j2\pi(\nu_x x + \nu_y y)} dx dy. \quad (1)$$

For a given near-field measurement, it can be shown via reciprocity that $\tilde{S}$ is related to the plane-wave spectrum of the probe, $\mathcal{E}_p$, and AUT, $\mathcal{E}_a$, as

$$\tilde{S}(\nu_x, \nu_y) \propto \nu_z \mathcal{E}_a(\nu_x, \nu_y) \cdot \mathcal{E}_p(\nu_x, -\nu_y) \quad (2)$$

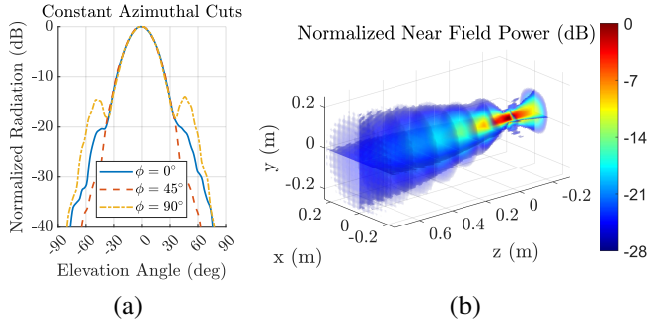


Figure 4: (a) Radiation pattern plotted for three cuts along a constant azimuth angle for the antenna depicted in Fig. 1. (b) 3D near-field visualization of the near-field amplitude both before and after the measurement plane ($z = 0$).

where $\nu_z = \sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2}$ [8]. The system reported in this work supports approximate probe compensation under the assumption that the field produced by the AUT in the measurement plane is linearly polarized along the vertical direction. Further, it is assumed that the probe is a rectangular waveguide probe with horizontal dimension a and vertical dimension b . Taking the $\hat{y}$ coordinate as the vertical direction and assuming that $\hat{x} \cdot \mathcal{E}_a \approx 0$ allows for $\mathcal{E}_a$ to be computed as

$$\mathcal{E}_{ay}(\nu_x, \nu_y) \approx \frac{\kappa \tilde{S}(\nu_x, \nu_y) / \nu_z \left(1 + \frac{\nu_y^2}{\nu_z^2}\right)}{\left[\text{sinc}\left(\pi \nu_x a - \frac{\pi}{2}\right) + \text{sinc}\left(\pi \nu_x a + \frac{\pi}{2}\right)\right] \text{sinc}(\pi \nu_y b)} \quad (3)$$

where κ is an arbitrary measurement constant. The probe-compensated spatial near-field as well as the $\hat{y}$ component of the plane-wave spectrum for the Narda 640 horn measured using the setup in Fig. 1 is plotted in Fig. 3. With the plane-wave spectrum computed, the far-field can then be obtained as [5]

$$|E_{\text{FF}}| \propto \frac{\sqrt{1 - \cos^2 \phi \sin^2 \theta}}{r} \left| \mathcal{E}_{ay} \left(\frac{\sin \theta \cos \phi}{\lambda}, \frac{\sin \theta \sin \phi}{\lambda} \right) \right|. \quad (4)$$

Note that this expression is simplified by assuming that the transverse component of the field in the aperture is entirely polarized along the $\hat{y}$ direction. The published open-source code [6] uses the far-field data and plane-wave spectrum as computed using (3) and (4) to generate insightful reports and radiation characteristics of the AUT. The plots in Fig. 4, for example, depict the far-field radiation pattern along various azimuth cuts and a 3D visualization of the near-field pattern.

C. Validation

Simulations carried out in HFSS to validate the presented near-field scanner are illustrated in Fig. 5, and offer three different metrics which can be used to benchmark the measured data. In “Simulation 1,” the Narda horn is simulated in isolation with a spherical radiation boundary to compute (i) the far-field pattern directly through the radiation integral and (ii) the near-field over a large aperture (6 meters) in the measurement plane. In “Simulation 2,” a waveguide probe is

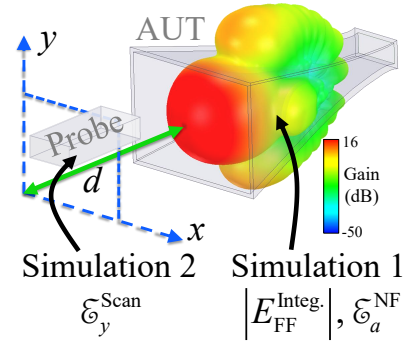


Figure 5: Illustration of the simulation setup used to validate the performance of the near-field scanner. In “Simulation 1” the far-field gain pattern of the AUT as well as the near-field in the measurement plane is captured. In “Simulation 2” the probe is included to compare the measured signals to the idealized setup.

introduced with the same separation as in the measurement and scanned within the measurement plane to simulate (iii) the value of S_{21} that is expected in measurement. The total far-field magnitude computed directly using the radiation integral within HFSS is referred to as $|E_{\text{FF}}^{\text{Integ.}}|$, the plane-wave spectrum computed using the near-field setup in HFSS is referred to as $\mathcal{E}_a^{\text{NF}}$, and the plane-wave spectrum computed using the simulated scan including both the AUT and the probe is referred to as $\mathcal{E}_{ay}^{\text{Scan}}$. These are all used to benchmark the measured probe-compensated measured near-field plane-wave spectrum $\mathcal{E}_{ay}^{\text{Meas}}$. The relative error between the measured data and the various HFSS simulations is reported in terms of the Root Mean Square Error (RMSE) in the spectral domain. In all cases, the error between a given field quantity, $F_{\vec{\nu}}$, and reference quantity, $R_{\vec{\nu}}$, will be calculated as

$$\text{Err}(F_{\vec{\nu}}, R_{\vec{\nu}}) \triangleq \sqrt{\sum_{\vec{\nu}} |F_{\vec{\nu}} - R_{\vec{\nu}}|^2} / \sqrt{\sum_{\vec{\nu}} |R_{\vec{\nu}}|^2}, \quad (5)$$

where $\sum_{\vec{\nu}} \{\cdot\}$ denotes a summation over all available spectral values and $\vec{\nu} = \nu_x \hat{x} + \nu_y \hat{y} + \nu_z \hat{z}$.

Perhaps the most relevant figure-of-merit for end-users of the presented system is the RMSE between (i) the measured and post-processed far-field pattern, $|E_{\text{FF}}^{\text{Meas.}}|$, and (ii) the simulated far-field pattern computed directly in HFSS using the radiation integral, $|E_{\text{FF}}^{\text{Integ.}}|$. This agreement can be seen in the E-/H-plane radiation patterns in Fig. 6, and the RMSE is $\text{Err}(|E_{\text{FF}}^{\text{Meas.}}|, |E_{\text{FF}}^{\text{Integ.}}|) = 0.0994$. A significant contributor to the RMSE is the truncation associated with the finite width of the measurement plane. This conclusion is supported by Fig. 7, which plots the RMSE between the far-field pattern computed using (i) simulated near-field data, $|E_{\text{FF}}^{\text{NF}}|$, and (ii) the radiation integral, $|E_{\text{FF}}^{\text{Integ.}}|$, for various measurement aperture sizes. The radiation pattern in the E-/H-plane for the near-field simulation using a sample area of $6\text{m} \times 6\text{m}$ is shown in Fig. 6. It can be seen that the larger scan width (6m as opposed to 0.5m) improves agreement between expected

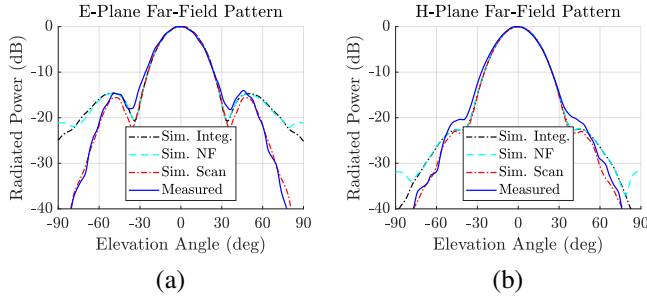


Figure 6: Normalized radiation patterns in the (a) E-Plane and (b) H-Plane as obtained in measurement (Measured) as well as three methods using HFSS simulation results. The ideal radiation pattern is calculated using the radiation integral in HFSS (Sim. Integ.). This is compared to the pattern computed from the near-field data given in HFSS (Sim. NF) and a simulated scan with both the horn and the probe (Sim. Scan).

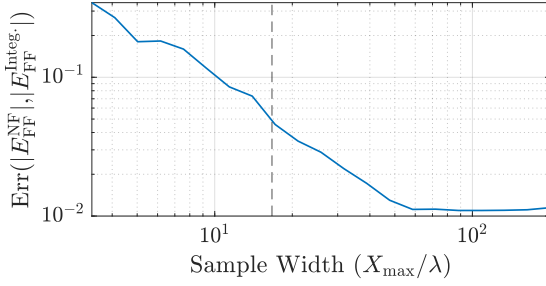


Figure 7: The RMSE between the far-field amplitude computed from the radiation integral and the near-field data versus the width of the near-field sample area. The black dashed line denotes the maximum scan width used in the measurement ($X_{\max} = 16.7\lambda$).

radiation pattern and the computed pattern from about 45° to 80° in elevation. The RMSE over the entirety of the visible region is $\text{Err}(|E_{\text{FF}}^{\text{NF}}|, |E_{\text{FF}}^{\text{Integ.}}|) = 0.0114$.

Another key functionality of the near-field scanner is to report the complex field values near a radiating source as shown in Fig. 3. This can be compared to the simulated near-field values for the vertical component of the plane-wave spectrum, $\mathcal{E}_{ay}^{\text{NF}}$ plotted in Fig. 8. While the amplitude and phase are visually similar to the measured values in Fig. 3, the RMSE is relatively high: $\text{Err}(\mathcal{E}_{ay}^{\text{Meas.}}, \mathcal{E}_{ay}^{\text{NF}}) = 0.2771$. This can be attributed to phase errors due to an imperfect measurement of the separation between the AUT and probe, unintended tilt between the AUT and the measurement plane, and accumulated position inaccuracies. To verify that the error is predominantly due to measurement error rather than algorithmic error, an ideal simulated near-field scan is carried out in HFSS (see “Simulation 2” in Fig. 5) and probe-compensated to obtain the plane wave spectrum. In this case, the RMSE was found to be $\text{Err}(\mathcal{E}_{ay}^{\text{Scan}}, \mathcal{E}_{ay}^{\text{NF}}) = 0.0761$ with the remaining error due to the approximate probe compensation, computational error in the HFSS simulations, and inherent approximations to

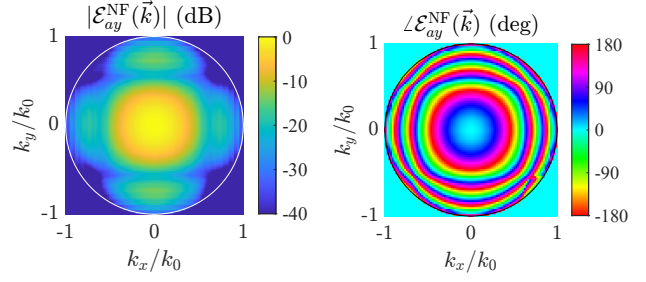


Figure 8: Plane-wave spectrum computed using the near-field samples from “Simulation 1” in Fig. 5. Note that the samples have been truncated to the same area as the measured scan area ($0.5\text{m} \times 0.5\text{m}$).

Table I: Table comparing the error between near- and far-field measurements and simulations.

Test	Reference	Err (Test, Reference.)
$ E_{\text{FF}}^{\text{Meas.}} $	$ E_{\text{FF}}^{\text{Integ.}} $	0.0994
$ E_{\text{FF}}^{\text{NF}} $	$ E_{\text{FF}}^{\text{Integ.}} $	0.0114
$\mathcal{E}_{ay}^{\text{Meas.}}$	$\mathcal{E}_{ay}^{\text{NF}}$	0.2771
$\mathcal{E}_{ay}^{\text{Scan}}$	$\mathcal{E}_{ay}^{\text{NF}}$	0.0761
$ \mathcal{E}_{ay}^{\text{Meas.}} $	$ \mathcal{E}_{ay}^{\text{NF}} $	0.0797

performing near-field measurements [10]. Further, comparing the amplitude of the measured plane-wave spectrum to that obtained using the simulated near-field data yields a lower RMSE: $\text{Err}(|\mathcal{E}_{ay}^{\text{Meas.}}|, |\mathcal{E}_{ay}^{\text{NF}}|) = 0.0797$. The results of this error study are summarized in Table I which reports the RMSE for all presented cases.

IV. PRACTICAL CONSIDERATIONS

Though the scanner does produce reliable results, there are some important systematic considerations. First, the theoretical maximum scan angle is limited by how close the probe can be to the AUT. This limit is dictated by the dimensions of the AUT and scanning plane [8]. To visualize this theoretical limit, Figure 9 plots the maximum scan angle of the system when the probe is separated from the AUT by $2D_p^2/\lambda$, where D_p is the largest dimension of the probe and λ is the free-space wavelength. This limit is given by

$$\theta_{\max} = \arctan\left(\frac{(X_{\max} - D_a)\lambda}{4D_p^2}\right), \quad (6)$$

where $X_{\max}$ is the maximum width the scanner can move relative to the probe and D_a is the largest dimension of the AUT aperture. A visual defining these variables can be found in Figure 10.

During measurements, a distance of 3λ to 5λ between the probe and AUT was used as recommended by [10]. This distance was greater than that of the far-field of the probe, leading to a measurement with a lower $\theta_{\max}$ than what is theoretically possible. The tradeoff for lower $\theta_{\max}$ was reliability; a greater distance minimizes the reactive energy captured in the measurement [11].

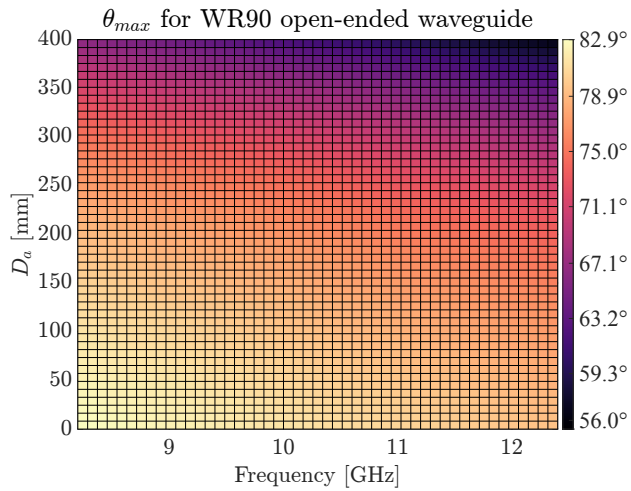


Figure 9: The theoretical greatest angle that can be measured with the planar near-field scanner, θ_{max} , plotted on a color ramp as a function of the AUT's aperture, D_a , and as a function of the frequency. Values calculated using equation (6) with a X_{max} value of 550 mm and a D_p value of about 25 mm. The frequency range shown and D_p selected are that of the WR90 open-ended waveguide probe used for testing.

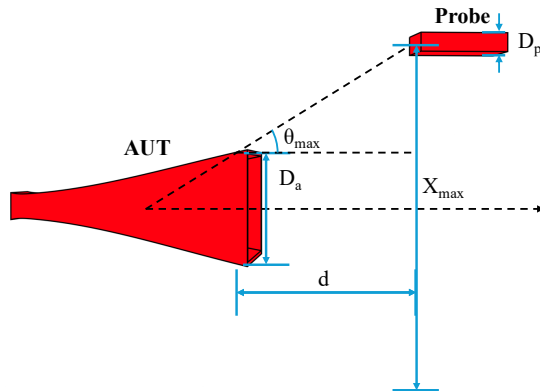


Figure 10: A physical depiction of the variables used in equation 6. X_{max} is the maximum scan width of the probe relative to the AUT, d is the distance between the AUT aperture and the probe, D_a is the width of the aperture, D_p is the greatest dimension of the probe, and θ_{max} is the largest angle relative to the AUT that can be measured by the probe. Inspired by Figure 5.2 from [10].

The design of the scanner was optimized around cost and simplicity rather than machine lifetime and precision. To improve spatial resolution of the mechanical assembly to sub-millimeter precision, the reader should explore other mature technology options such as linear guides and ball screws. In addition, the near-field scanner has only been tested with single polarization X band horns and probes. Higher frequencies and dual-polarized measurements are hypothetically possible but untested. Finally, and most importantly, the resources and documentation provided in [6] are unverified; hypothetically,

a reader should be able to reconstruct the planar near-field scanner presented in this paper, but critical components of the design may be missing from [6], and contact with the repository maintainers may be required.

V. CONCLUSION

The planar near-field scanner presented in this paper improves the accessibility of near-field scanning, particularly for hobbyists, research groups, and small businesses. Though confined to high directivity radiators for full characterization of radiation patterns due to the limitations of planar near-field scanners, the reliable measurement from 45 degrees off boresight provides quick and easy insight into radiation patterns. The open-source hardware and software used to build and run the machine provides a resource for future custom planar near-field scanners and a starting point for any measurement setup dependent on large scale linear actuation. Future improvements and upgrades to the scanner can push the scanner's capabilities beyond what is outlined in this paper, allowing for tailored measurements of unique electromagnetic devices and materials.

ACKNOWLEDGMENT

The authors thank Dr. Songyi Yen for advice and support, especially for the theoretical aspects of the scanner.

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